DINESH SAMIAH PILLAI

AI PRODUCT MANAGEMENT | AI-ASSISTED LEGACY MODERNIZATION | INSURANCE SOLUTION ARCHITECTURE

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PROFESSIONAL PROFILE

Technical product and solution leader with 20 years in enterprise software, including 14 years in property & casualty insurance. Led a 10-person team that used Anthropic Claude to modernize a 500+ program COBOL/mainframe commercial lines platform into Python microservices on AWS, live in production in 4–6 months. Combines product ownership (business case, stakeholder alignment, success metrics, phased rollout) with hands-on cloud, data, and insurance-domain architecture. AWS Certified Solutions Architect with AI/ML Product Management and Generative AI training.

SIGNATURE PROJECT

AI-Assisted Commercial Lines Modernization | Mainframe COBOL to Python, Anthropic Claude, AWS, BigQuery

500+ COBOL programs • **10**-person team • **4** states (CA, OR, WA, ID) • Live in production in **4–6 months** • **$1M+** premium accounts

- **Product leadership:** Built the business case, secured stakeholder approval, defined success metrics, and planned the phased rollout to replace the mainframe commercial lines platform.
- **AI-assisted engineering:** Applied Anthropic Claude across the full lifecycle: COBOL comprehension and documentation, business-rule extraction, Python code generation, and test generation, with human review and engineering guardrails on every change.
- **Rating engine:** Rebuilt every rating module across commercial lines of business, including complex experience rating, General Liability, and Property, with state-specific rules for CA, OR, WA, and ID.
- **Data-driven rates:** Moved rate tables and rating factors out of hard-coded COBOL into Google BigQuery, so rates are maintained as data rather than code.
- **Forms flagging:** Re-engineered policy forms and endorsement flagging rules into a JSON-based rules engine, replacing logic embedded in COBOL.
- **Validation:** Combined AI agent-driven QA with premium reconciliation against legacy-system rating worksheets to prove rating parity before go-live.
- **Platform scope:** Delivered as AWS API-based microservices, including data migration, commissions, and financial reporting, supporting large commercial accounts.

PROFESSIONAL EXPERIENCE

MPHASIS CORPORATION | Technical Lead / Project Manager / Solution Architect | Aug 2008 – Present

Key client: Oregon Mutual Insurance (2012 – Present)

- Led the Anthropic Claude-assisted modernization of the commercial lines mainframe platform to Python microservices on AWS (see Signature Project).
- Lead technical analysis, solution design, and delivery coordination for insurance platforms spanning claims, billing, policy administration, commercial lines, and reporting.
- Architected a resilient AWS serverless OFAC screening integration using EventBridge and Step Functions, removing the need for a costly third-party application upgrade.
- Designed a cloud-based claims letter automation solution using AWS RDS, APIs, and stored procedures, resolving browser-related security concerns and improving performance.
- Advanced cloud modernization with AWS VPC, SSO, Terraform, Glue, Athena, CloudTrail, EMR, and organization-wide security policies.
- Partner with business stakeholders, engineering, and vendors on architecture trade-offs, feasibility, release readiness, and issue resolution.

UST GLOBAL | Software Engineer | Aug 2006 – Aug 2008

- Evaluated technical solutions against customer requirements, budgets, and product goals, working with engineers and third-party product owners.
- Built prototypes and application interfaces supporting product development within Agile engineering practices.

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KEY ARCHITECTURE & DELIVERY PROJECTS

OFAC Screening Replacement | AWS EventBridge, Step Functions, Serverless

- **Challenge:** Continuing OFAC compliance screening would have required a costly upgrade of a third-party application.
- **Contribution:** Designed and supported deployment of a resilient, event-driven serverless integration, with EventBridge for event routing and Step Functions for orchestration.
- **Outcome:** Kept compliance screening in place while avoiding the third-party upgrade cost.

Claims Letter Automation | AWS RDS, APIs, Stored Procedures

- **Challenge:** The existing claims letter process raised browser-related security concerns and had performance problems.
- **Contribution:** Designed a cloud-based letter application, migrated letter data to AWS RDS, and specified APIs that call stored procedures to generate letters.
- **Outcome:** Resolved the security concerns and improved application performance.

Reporting ETL Migration & Data Archival | AWS EMR, S3/Glacier, Glue, Athena

- Played a key role in migrating legacy IBM Cognos ETL workloads to AWS EMR for greater scalability and processing efficiency.
- Designed a legacy data archival approach on S3/Glacier with lifecycle policies and cost estimates to meet insurance record-retention requirements.

Insurance VIN Lookup | Socket Integration

- Designed and developed a socket-based vehicle lookup integration for insurance online modules, improving response time.

AIG VALIC Fee Disclosure | Compliance Delivery

- Prepared high- and low-level designs; coordinated statements of work, stakeholder approvals, development, code review, QA, and deployment.

CORE CAPABILITIES

AI & Modernization: Anthropic Claude; AI-assisted COBOL-to-Python conversion; AI-driven business-rule extraction; AI-generated code and tests; AI agent-driven QA; Generative AI; AI/ML product management

Product & Delivery: Business cases; success metrics; phased rollout planning; stakeholder alignment; technical requirements; feasibility and cost-benefit analysis; team leadership; Agile/Kanban; Jira; Confluence

Cloud & Data: AWS (API-based microservices, S3, EC2, Lambda, IAM, RDS, EventBridge, Step Functions, EMR, Glue, Athena, VPC); Google BigQuery; JSON rules engines; Terraform; Python; SQL Server; DB2; SSIS/SSRS

Insurance: Commercial lines P&C; rating (experience rating, General Liability, Property); forms and endorsement rules; policy administration; claims; billing; commissions; financial and regulatory reporting; data migration

CERTIFICATIONS & EDUCATION

AWS Certified Solutions Architect – Associate (2018) • AI/ML Product Management – ELVTR • Generative AI Fundamentals – Databricks • IBM Certified Database Associate – DB2 (2008)

Bachelor of Engineering, Computer Science | Maharaja Engineering College, Anna University, India | 2006

ADDITIONAL TECHNOLOGIES

COBOL, JCL, IMS DB/DC, CICS, VSAM, GitHub, Azure DevOps, ServiceNow, Linux, Unix shell scripting, PowerShell, CrushFTP